Open Engineering Architecture & Agentic Systems Series

# Spec-Driven Development, Spec-as-SSOT, Conductor & OpenSpec

## The Concise Practical Field Guide: Workflows, Commands, Playbooks & Copy-Pasteable Templates

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**What Is Inside This Concise Field Guide (5 Focused Sections)**

- **Section 1 — Why SDD, Spec-as-SSOT & 5-Minute Quickstart**: Escaping "Vibe Coding", the 5-minute install & init commands, Spec-First vs. Spec-Anchored vs. Spec-as-SSOT, RFC 2119 (`SHALL`/`MUST`) + Gherkin (`WHEN`/`THEN`), and when to use Conductor vs. OpenSpec vs. `SPEC.md`.
- **Section 2 — Conductor Hands-On Guide ("Measure Twice, Code Once")**: The 5 core capabilities, `conductor/` directory layout, the 3-step daily loop (`/conductor:setup` → `/conductor:newTrack` → `/conductor:implement`), Red-Green-Refactor TDD with `git notes` and `[checkpoint: <sha>]`, and Day-2 commands (`status`, `review`, `revert`).
- **Section 3 — OpenSpec Hands-On Guide (The "Core 4" & Delta Specs)**: Two-plane directory layout (`openspec/specs/` vs. `openspec/changes/`), `/openspec-*` vs. `/opsx:*` naming, the 99% vs. 1% CLI rule, the "Core 4" workflow (`explore` → `propose` → `apply` → `archive`), and a complete change bundle (`ADDED`, `MODIFIED`, `REMOVED`, `RENAMED`).
- **Section 4 — OpenSpec Playbooks, Stacked PRs (`git` / `jj`) & Security Guardrails**: The 4 scenario playbooks, mapping `1 Change = 1 Feature` across Single-PR, Two-Stage Spec-First PR, and Stacked Multi-PRs (`tasks.md` Batches + Dedicated Archive PR `N+1`), and the 5 mandatory Golden Rules.
- **Section 5 — Spec-as-SSOT (`SPEC.md`) Rebuild Test & All 11 OpenSpec Skills Reference**: The 8-section `SPEC.md` blueprint, 3-Phase Clean-Room Rebuild Test, Cognitive Layer vs. Deterministic CLI Engine architecture, and the complete 11-skill reference table with session outputs.

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# 1 Why SDD, Spec-as-SSOT & 5-Minute Quickstart

**Why Spec-Driven Development? (Escaping "Vibe Coding").** When you ask an AI coding assistant in a chat window to *"add webhook retries"* or *"refactor the auth middleware"* without a written specification, the agent guesses your architecture from whatever files happen to be in its context window. In production repositories, unconstrained conversational prompting (**"Vibe Coding"**) consistently breaks down in four ways illustrated in Figure 1 below:

- **Silent Architectural Drift:** The agent invents new database tables, installs unapproved packages, or bypasses existing service layers because no file defines your repository's boundaries.
- **Unstated Edge Cases:** Chat prompts describe the happy path; rate limits, replay attacks, and rollback behavior are left to the model's imagination.
- **Painful Code Reviews:** Teammates are forced to reverse-engineer what you *intended* to build by reading 1,500 lines of AI-generated diffs.
- **Context Amnesia:** The moment you close your terminal or start a new chat session, the architectural rationale behind yesterday's prompt vanishes.

**Specificational Coding vs. Conversational Coding & Ephemeral Plan Mode.** **Specificational coding (Spec-Driven Development)** is where an engineer collaborates with an AI agent to refine a persistent **Specification** (`spec.md`), and the agent then writes the **code** and **tests** from that contract. Unlike **conversational coding ("Vibe Coding")** (where you chat back-and-forth to tweak source code directly) and unlike **ephemeral planning mode** (which creates a disposable task checklist for a single turn), a specification is scoped to a module or service and stays in Git so you see how every change maps to existing API contracts. **The ultimate goal of specificational coding is to make source code a compiled artifact of the spec, just like machine code is a compiled artifact of source code today.** Once you and the agent agree on the spec, all work flows through **three core spec actions**:

1. **Apply (`Act on Spec` — Labs 01, 03, 06):** Convert approved requirements (`REQ-0001..0004`) into production code and 1-to-1 unit tests (`/conductor:implement`, `/opsx:apply`).
2. **Sync (`Sync Spec` — Labs 02, 04, 05):** Inspect existing brownfield code or spec revisions and synchronize `spec.md` *without* mutating working code (`/opsx:explore`, `/opsx:sync`).
3. **Clean / Archive (`Clean Spec` — Lab 05):** Strip transient checklists and merge verified deltas into the permanent specification (`/opsx:archive`).

**5-Minute Terminal Quickstart (Unified `TaskFlow` App).** You can install and initialize both **Conductor** and **OpenSpec** on the `TaskFlow` personal productivity app in under two minutes:

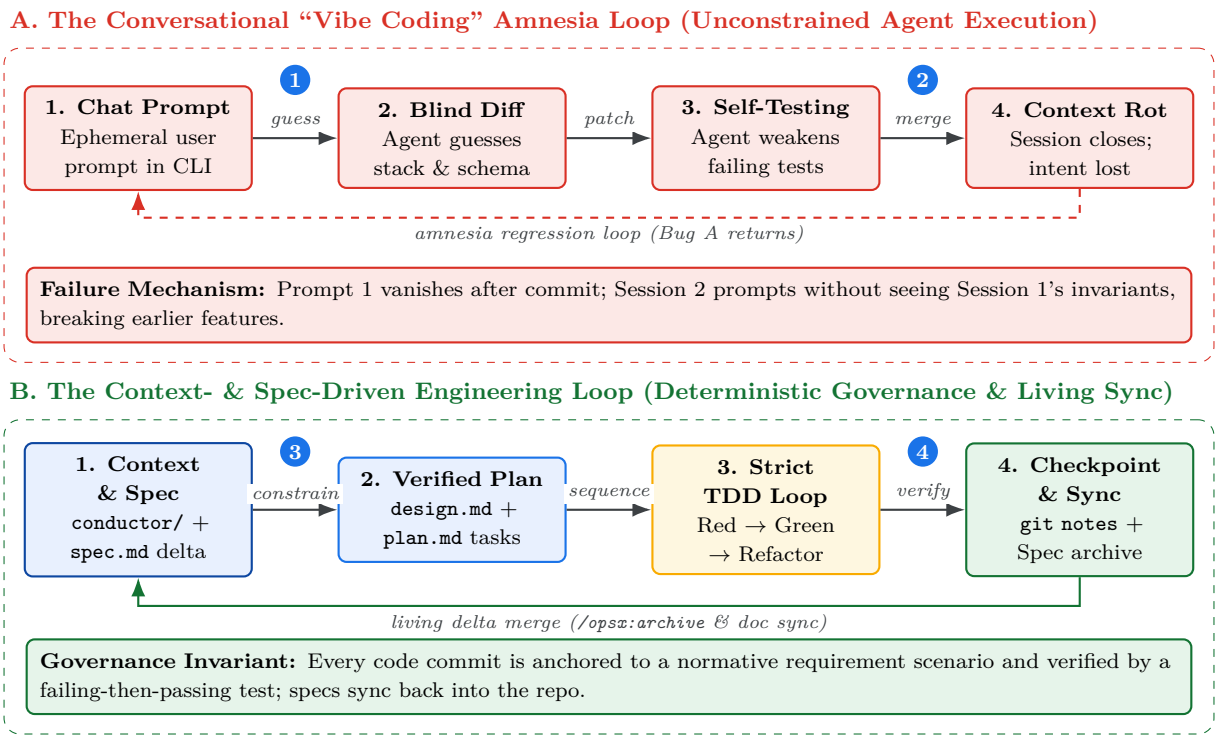



Figure 1: Contrast between the "Vibe Coding" Amnesia Loop (A, top: Pins [1] ephemeral prompt and [2] self-testing trap) and the Context- and Spec-Driven Engineering Loop (B, bottom: Pins [3] repository spec lock and [4] verified checkpoint archive).

**5-Minute Terminal Quickstart: Install & Initialize Conductor and OpenSpec on `TaskFlow`**

```
# 1. Install Conductor extension for Gemini CLI & OpenSpec CLI globally
$ gemini extensions install https://github.com/gemini-cli-extensions/conductor --auto-update
$ npm install -g @fission-ai/openspec@latest

# 2. Initialize OpenSpec in your repository
$ cd ~/projects/taskflow && openspec init
# Creates openspec/config.yaml, openspec/specs/, and openspec/changes/

# 3. Launch Gemini CLI and initialize Conductor repository context
$ gemini
> /conductor:setup
# Generates conductor/{index,product,product-guidelines,tech-stack,workflow,tracks}.md

# 4. Propose a capability spec delta (OpenSpec) & run a TDD feature track (Conductor)
> /opsx:propose "Add core TodoList with add_task, complete_task, and delete_task"
> /conductor:newTrack "Implement TaskFlow TodoList (REQ-0001..REQ-0004)"
> /conductor:implement
```

## 1.1 The 3 Levels of SDD Maturity & Choosing the Right Tool

**Spec-First vs. Spec-Anchored vs. Spec-as-Source (SSOT).** Teams adopt Spec-Driven Development across three practical maturity levels summarized in Table 1:

1. **Level 1 — Spec-First (Disposable Planning):** You write a spec or plan before coding a feature, use it to guide the AI agent for that single PR, and then let the spec go stale. Useful for one-off spikes, but

future sessions still suffer from context rot.

2. **Level 2 — Spec-Anchored (Living Repository Specs):** Specifications live permanently in Git alongside your code (`openspec/specs/` and `conductor/`). Every feature branch updates the spec delta and archives it back into the main spec tree upon completion so specs and code never drift.
3. **Level 3 — Spec-as-Single-Source-of-Truth (Spec-as-SSOT / `SPEC.md`):** An 8-section canonical `SPEC.md` file captures all essential system behavior (domain intent, data schemas, API contracts, state machines, and parameterized Terraform) with enough precision to pass **The Rebuild Test**—regenerating a working, test-passing service from `SPEC.md` alone in an empty directory.

| Tool / Paradigm | Files on Disk | Core Strength | When to Use It |
|---|---|---|---|
| **Conductor** (`conductor/`) | `product.md`, `tech-stack.md`, `workflow.md`, `tracks/<id>/{spec,plan}.md` | Project-wide context lock, strict Red-Green-Refactor TDD (>80% cov), 7-char commit SHAs, `git notes`, `/conductor:revert`. | Daily feature & bug implementation where strict TDD, stack governance, and clean Git rollback matter most. |
| **OpenSpec** (`openspec/`) | `specs/<cap>/spec.md` + `changes/<name>/{proposal, specs, design, tasks}` | Two-plane delta specs (`ADDED`, `MODIFIED`, `REMOVED`, `RENAMED`), RFC 2119 + Gherkin validation (`validate --strict`). | Features touching domain APIs/schemas where reviewing a crisp behavioral spec diff before coding is critical. |
| **Spec-as-SSOT** (`SPEC.md`) | Root 8-Section `SPEC.md` blueprint | Complete system DDL, API contracts, parameterized IaC, and 3-Phase Clean-Room Rebuild Test. | Extracting a reusable blueprint from a brownfield repo, cross-language rewrites, or greenfield service design. |

Table 1: Practical comparison of Conductor, OpenSpec, and Spec-as-SSOT (`SPEC.md`).

**Why Specifications Use ALL-CAPS (`SHALL`, `MUST`) and Gherkin (`WHEN` / `THEN`).** Both OpenSpec and `SPEC.md` use two industry standards to eliminate natural-language ambiguity:

- **IETF RFC 2119 & RFC 8174 ALL-CAPS Keywords (`SHALL`, `MUST`, `SHOULD`, `MAY`):** Casual prose (*"returns 429 on rate limit"*) is ambiguous. Following RFC 2119 and RFC 8174 (which specifies that keywords are normative *only* in uppercase), `SHALL` and `MUST` define hard invariants that fail verification if violated, `SHOULD` defines recommended defaults requiring justification in `design.md` to bypass, and `MAY` defines optional behavior.
- **Gherkin Scenarios (`GIVEN` / `WHEN` / `THEN` / `AND`):** Every requirement is paired with concrete `#### Scenario:` blocks that map 1-to-1 to unit/integration tests: `GIVEN` sets up test fixtures, `WHEN` triggers the action, and `THEN` asserts the exact HTTP status, database state, or output payload.

**The Generation–Verification Loop, Human Interaction Gates & Karpathy's 4 Agent Rules.** In **Software 3.0**, code generation is virtually free while **human verification is the true bottleneck**. To keep coding agents on a "short leash" (the **Autonomy Slider**), Conductor and OpenSpec enforce four behavioral rules and an immutable verification harness (mirroring Karpathy's `autoresearch` separation between human-steered `program.md` and read-only evaluator `prepare.py`):

1. **Think Before Coding (Plan Mode Q&A Gate):** Never guess silently on ambiguous requirements; pause in `/conductor:newTrack` or `/opsx:explore` to surface trade-offs and ask 3–5 clarifying questions before touching code.
2. **Simplicity First (Spec-First Diff Review):** Humans review and approve a 50-line behavioral contract (`spec.md` + ADRs) before implementation, and the agent writes only the minimal code needed to turn

failing tests green.

3. **Surgical Changes (Out-of-Scope & Tech-Stack Locks):** Lock non-target findings into `## 6. Out of Scope` and halt on any unapproved dependency not listed in `conductor/tech-stack.md`.
4. **Goal-Driven Execution & Immutable Verifiers (`chmod -w`):** Anchor every task to a 1-to-1 `test_reqXXXX_*` unit test and phase checkpoint, and lock adversarial verification suites read-only (`chmod -w`) during Clean-Room Rebuild Tests so an agent can never pass by weakening test assertions.

**Joel Spolsky's Law of Leaky Abstractions (Why Code Generation Saves Time *Working*, Not Time *Learning*).** In his classic 2002 essay *"The Law of Leaky Abstractions"* (*Joel on Software*), Joel Spolsky formulated the foundational law that governs every high-level abstraction—and anticipates the central trap of conversational "Vibe Coding" with AI agents:

**System Design Invariant & Tie-Breaker**

Joel Spolsky's Law of Leaky Abstractions (2002) Applied to AI Code Generation **"All non-trivial abstractions, to some degree, are leaky."**

*"The law of leaky abstractions means that whenever somebody comes up with a wizzy new code-generation tool that is supposed to make us all ever-so-efficient, you hear a lot of people saying 'learn how to do it manually first, then use the wizzy tool to save time.' Code generation tools which pretend to abstract out something, like all abstractions, leak, and the only way to deal with the leaks competently is to learn about how the abstractions work and what they are abstracting.* ***So the abstractions save us time working, but they don't save us time learning.*** *"* — **Joel Spolsky**, *Joel on Software* (Nov 11, 2002)

**Engineering Consequence for SDD:** Natural-language prompts are a leaky abstraction even on everyday apps like `TaskFlow`. Prompting *"build a to-do list"* or *"don't exceed my daily minute budget"* hides edge cases: should blank titles (`" "`) be allowed? If 60 minutes remain in a 120-minute daily budget, should a 60-minute task succeed (`<=`) or fail (`<`)? **Spec-Driven Development** forces you to explicitly specify and test the exact boundary where the abstraction leaks (`REQ-0001..REQ-0004`, `ADR-*.md`, and `#### Scenario:`) before the agent generates code.

**Common Misconception vs. Engineering Reality: Why Ephemeral Chat Plan Mode Is Not Spec-Driven Development**

× **Common Misconception:** *"Toggling Plan Mode in my coding CLI for a single turn gives me all the benefits of Spec-Driven Development."* ✓ **Engineering Reality:** Ephemeral chat plans are Level-1 disposable checklists that vanish when the terminal closes, leaving the next session blind to why edge cases were chosen. Spec-Driven Development (Level 2 *Spec-Anchored* and Level 3 *Spec-as-SSOT*) persists `openspec/specs/` and `conductor/` in Git so every future feature diffs against living domain contracts.

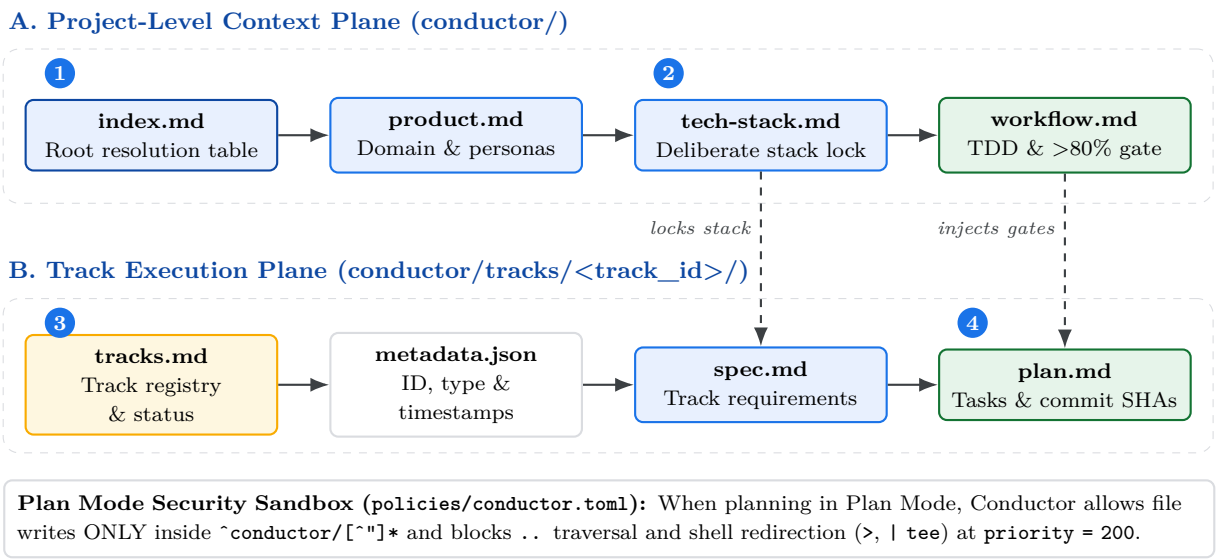
**Stop and Think: Why Must Adversarial Rebuild Tests Be Locked Read-Only (`chmod -w`)?**

**Question:** During a Software 3.0 implementation loop, why is it dangerous to give the coding agent write permissions to the acceptance test suite (`adversarial_tests/`)? **Answer:** Because an unconstrained LLM optimizes for passing the test runner via the path of least resistance—often weakening assertions or deleting failing edge-case tests rather than fixing the underlying implementation bug. Locking verifiers read-only (`chmod -R a-w`) forces the implementation to conform to the specification contract.

# 2 Conductor Hands-On Guide: "Measure Twice, Code Once"

**Control Your Code Through Persistent Repository Context.** Conductor is built on a simple engineering principle: *"Measure twice, code once—failing to plan is planning to fail."* Instead of re-explaining your tech stack, coding conventions, and testing rules in every chat session, Conductor stores project context as version-controlled Markdown artifacts inside `conductor/` (Figure 2 below), unlocking five practical capabilities:

1. **Plan before you build:** Creates explicit specifications (`spec.md`) and phased task plans (`plan.md`) for your approval before any application code is written.
2. **Maintain context:** Automatically enforces your style guides (`conductor/code_styleguides/`), technology choices (`tech-stack.md`), and workflow rules (`workflow.md`) across every session.
3. **Iterate safely:** Lets you inspect and edit concrete Markdown plans before a single source file is touched.
4. **Work as a team:** Commits `conductor/` to Git so every developer and agent session shares the exact same project foundation.
5. **Build on existing ("brownfield") projects:** Automatically scans mature repositories (`pyproject.toml`, `package.json`, `go.mod`) while respecting `.gitignore` and `.geminiignore`.




**Plan Mode Security Sandbox (`policies/conductor.toml`):** When planning in Plan Mode, Conductor allows file writes ONLY inside `^conductor/[^"]*` and blocks `..` traversal and shell redirection (`>`, `| tee`) at `priority = 200`.

Figure 2: Conductor's two-plane directory architecture separating persistent project context (Plane A: Pins [1] root index and [2] stack lock) from feature track execution artifacts (Plane B: Pins [3] track registry and [4] TDD plan checkpoints).

## 2.1 The 3-Step Daily Workflow (`setup` → `newTrack` → `implement`)

**Step 1: Establish Context (`/conductor:setup`).** Run `/conductor:setup` once per repository. Conductor detects whether the repo is greenfield or brownfield, asks at most 4 interactive questions per phase (distinguishing **Additive** multi-select scope questions from **Exclusive** single-select architecture questions), and generates `conductor/{index,product,product-guidelines,tech-stack,workflow,tracks}.md`. If interrupted, re-running `/conductor:setup` automatically resumes at the exact incomplete file without overwriting prior work.

**Step 2: Specify and Plan (`/conductor:newTrack`).** Run `/conductor:newTrack "<description>"` whenever you start a feature or bug fix. Conductor reads `product.md` and `tech-stack.md`, asks brief clarifying questions,

and generates a track folder (`conductor/tracks/<track_id>/`) containing `spec.md` (requirements and acceptance criteria), `plan.md` (phased task checklist with automatic phase verification tasks), and `metadata.json`, then pauses for your review.

**Step 3: Implement with Strict TDD (`/conductor:implement`).** Once you approve `plan.md`, run `/conductor:implement`. Conductor steps through each task sequentially using strict Red-Green-Refactor TDD:

1. Marks the active task `[~]` in `plan.md`.
2. **Red Phase:** Writes failing unit/contract tests first and runs the test suite (`CI=true`) to confirm failure.
3. **Green & Refactor Phase:** Writes the minimal production code to pass tests and verifies `>80%` coverage.
4. **Commit & Audit Trail:** Commits the code, attaches a structured rationale via `git notes add -m "..." <sha>`, and marks the task `[x] <7-char-sha>` in `plan.md`.
5. **Phase Checkpoint:** At the end of each phase, runs the full test suite across all modified files, pauses for your manual verification approval, and records `[checkpoint: <7-char-sha>]` on the phase header.

**Hands-On Walkthrough: Conductor `spec.md`, `plan.md` & Terminal Execution Trace (`TaskFlow`)**

```
# === 1. Start a New Feature Track on TaskFlow (Lab 01 TodoList) ===
> /conductor:newTrack "Implement TaskFlow Core TodoList (REQ-0001..REQ-0004)"
# Creates conductor/tracks/todo_list_mvp/{spec.md, plan.md, metadata.json}

# === 2. Generated conductor/tracks/todo_list_mvp/spec.md (Excerpt) ===
# Track Specification: TaskFlow Core To-Do List (`todo_list_mvp`)
## 3. Functional Requirements
- REQ-0001: `add_task(title)` trims whitespace, assigns sequential `id` (1, 2, ...),
  sets `completed=False`, and raises `ValueError` on blank titles.
- REQ-0002: `complete_task(task_id)` sets `completed=True` or raises `KeyError`.
- REQ-0003: `get_pending()` and `get_completed()` filter tasks in creation order.
- REQ-0004: `delete_task(task_id)` deletes the task and returns `True` or `False`.

# === 3. Generated conductor/tracks/todo_list_mvp/plan.md (During Execution) ===
# Implementation Plan: TaskFlow Core To-Do List
## Phase 1: Core Creation & Completion [checkpoint: 9f8e7d6]
- [x] Task 1.1: Implement `add_task` and blank-title validation a1b2c3d
  - [x] Write failing `test_req0001_add_task_trims_title_and_rejects_blank`
  - [x] Implement `add_task` in `todo_list.py`
- [x] Task: Conductor - User Manual Verification 'Phase 1' 9f8e7d6

## Phase 2: Filtering & Deletion
- [~] Task 2.1: Implement `get_pending`, `get_completed`, and `delete_task`
  - [x] Write failing `test_req0003_*` and `test_req0004_*` unit tests
  - [ ] Implement filtering and deletion methods in `todo_list.py`

# === 4. Inspecting the Attached Git Note for Task 1.1 ===
$ git notes show a1b2c3d
Task 1.1: Implement add_task and blank-title validation
- Summary: Trims title, assigns sequential integer ID starting at 1, rejects empty strings.
- Files: todo_list.py, test_todo_list.py
- Verification: CI=true python3 -m unittest (4 passed, 100% coverage)
```

## 2.2 Day-2 Operations: Stack Lock, `/conductor:status`, `review` & `revert`

**Tech-Stack Deviation Gate, Code Review & Logical Git Revert.** Table 2 summarizes all six Conductor commands and how they handle mid-track stack changes, automated code reviews, and logical rollbacks.

| Command / Rule | How It Works | Practical Behavior & Guardrails |
|---|---|---|
| `/conductor:setup` | Scans repo (respecting `.gitignore`/`.geminiignore`) & interviews user. | Creates `conductor/*.md`. Resumes automatically across 7 state tiers if interrupted. |
| `/conductor:newTrack` | Interactive Q&A → generates `spec.md`, `plan.md`, `metadata.json`. | Injects a mandatory `User Manual Verification` task at the end of every phase in `plan.md`. |
| `/conductor:implement` | Executes `[ ]` → `[~]` → `[x] <sha>` via Red-Green-Refactor TDD. | **Stack Lock Gate:** If a task requires a new library not in `tech-stack.md`, the agent MUST halt, ask permission, and update `tech-stack.md` first. |
| `/conductor:status` | Reads `tracks.md` and active `plan.md` files. | Prints completed `[x]`, in-progress `[~]`, pending `[ ]` counts, and next actionable task. |
| `/conductor:review` | Audits diff against `spec.md`, `tech-stack.md`, and `code_styleguides/`. | Uses unified diff if `<300` lines or file-by-file chunking if `>300` lines; can auto-apply fixes into a new `## Phase: Review Fixes` block in `plan.md`. |
| `/conductor:revert` | 4-phase logical Git revert of a Track, Phase, or Task. | Collects implementation SHAs + `conductor(plan)` SHAs from `plan.md` (finding rebased "ghost commits" via commit message search) and runs `git revert --no-edit` in reverse order. |

Table 2: Complete reference for Conductor slash commands and execution guardrails.

## 2.3 Dual-Engine Wiring: Combining Conductor with Dedicated Spec Skills & OpenSpec

**Living Guardrails & The 5-Step Brownfield Refactoring Lifecycle.** You never need to re-run `/conductor:setup` to add new team rules or wire OpenSpec and custom spec skills into Conductor: because `conductor/index.md` and `conductor/workflow.md` are re-read at the start of every track creation, implementation, and review cycle, editing them in Git immediately updates agent behavior across your entire team. In production brownfield repositories, combine **Context-Driven Development (Conductor)** and **Spec-Driven Development (OpenSpec / Dedicated Spec Skill)** via the 5-step refactoring loop and two-file handshake shown below:

**Dual-Engine Handshake (`conductor/workflow.md` + `index.md`) & 5-Step Refactoring Loop**

```
# === 1. Add to conductor/workflow.md (Track Specification Policy) ===
### Track Specification Policy (Dual-Engine CDD + SDD Handshake)
- When executing `/conductor:newTrack`, conduct interactive Q&A to clarify scope.
- Upon questioning completion, invoke `@writing-arch-specs`
  (`.agents/skills/writing-arch-specs/SKILL.md`) and align with `openspec/changes/`
  to draft `conductor/tracks/<track_id>/spec.md` using the 7-Section Anatomy:
  1. Overview
  2. Architecture & Component Topology
  3. Functional Requirements
  4. Non-Functional Requirements
```

```
  5. Acceptance Criteria
  6. Out of Scope (locks non-critical scorecard items against scope creep)
  7. Verification Commands

# === 2. Expose in conductor/index.md ===
## Capabilities & Spec Contracts
- [Writing Architecture Specs Skill](../.agents/skills/writing-arch-specs/SKILL.md)
- [OpenSpec Capability Contracts](../openspec/specs/)

# === 3. The 5-Step Brownfield Refactoring Workflow (Lab 04 DailyPlanner) ===
Step 1: Review Scorecard Findings (`scorecard.md`: F-01 `>=` bug vs. F-02..F-04 wishes)
Step 2: Lock Scope (`F-01` = FIX NOW; `F-02..F-04` = DEFER into `## 6. Out of Scope`)
Step 3: Draft 7-Section `spec.md` & TDD `plan.md` via `/conductor:newTrack`
Step 4: Phased TDD Implementation (`/conductor:implement`: Red test -> Fix `>=` to `>`)
Step 5: Multi-Axis Audit (`/conductor:review` + `/opsx:verify`) -> `/opsx:archive`
```

**Common Misconception vs. Engineering Reality: Why You Never Need to Re-Run `/conductor:setup` to Update Team Rules**

× **Common Misconception:** *"Whenever our team adopts a new testing rule, linter, or spec skill, we must re-run `/conductor:setup` to regenerate the `conductor/` directory."* ✓ **Engineering Reality:** Conductor dynamically re-reads `conductor/index.md`, `workflow.md`, and `tech-stack.md` at the start of every `newTrack`, `implement`, and `review` command. Editing those Markdown files in Git immediately enforces the updated guardrails across every teammate's next session.

**Stop and Think: How Does `/conductor:revert` Recover After a Git Rebase?**

**Question:** Suppose `/conductor:implement` records 7-character commit SHAs (`a1b2c3d`) in `plan.md`, and you subsequently rebase your branch onto `main` (changing every commit hash). Why doesn't `/conductor:revert` fail with "unknown revision"? **Answer:** Conductor's 4-phase logical revert treats recorded SHAs as primary pointers but falls back to searching the Git log by task commit message and attached `git notes` to resolve rebased "ghost commits" before running `git revert --no-edit` in reverse topological order.

# 3 OpenSpec Hands-On Guide: The "Core 4" & Delta Specifications

**How OpenSpec Organizes Your Repository (Two-Plane Directory Model).** When you run `openspec init`, OpenSpec creates an `openspec/` directory that separates **Current System Truth (Plane 1: `openspec/specs/`)** from **Active Change Proposals (Plane 2: `openspec/changes/`)**. Instead of editing master specifications directly while coding, every feature or bug fix lives in its own isolated folder (`openspec/changes/<change-name>/`) containing four Markdown files—`proposal.md`, `specs/<domain>/spec.md`, `design.md`, and `tasks.md`—and is merged into `openspec/specs/` only when you run `/openspec-archive-change` (Figure 3 below). In a **multi-project monorepo**, run `openspec init --force` inside the specific sub-package directory (e.g., `services/order-api/openspec/`) rather than at the monorepo root so `openspec/specs/` stays tightly scoped to that service.

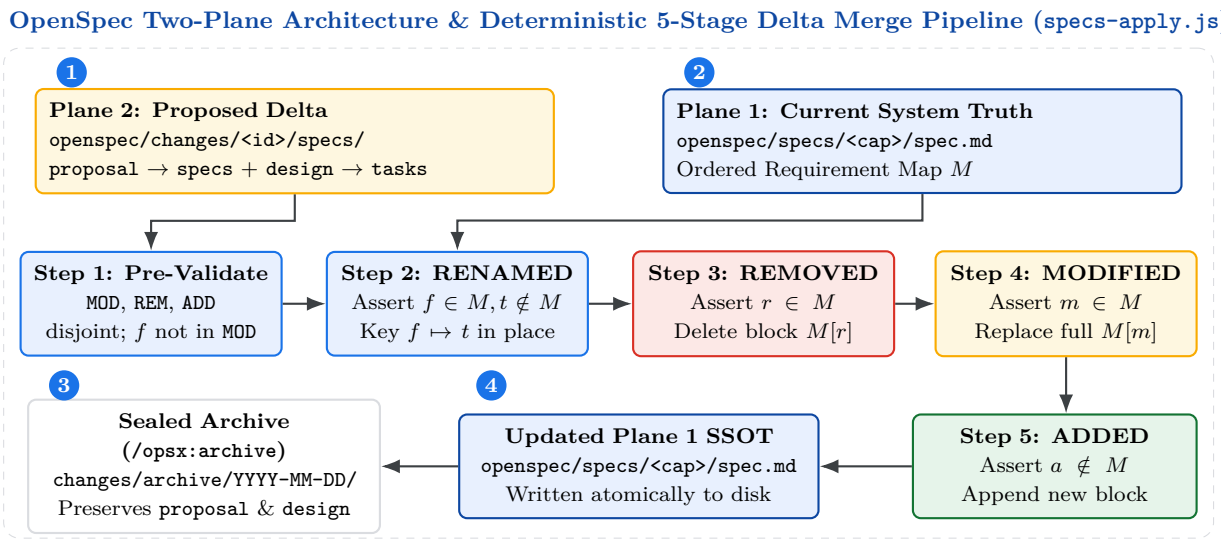



Figure 3: OpenSpec's Two-Plane directory model (Pins [1] Plane 2 proposed delta and [2] Plane 1 system truth) and 5-stage delta merge pipeline (`Pre-Validate` → `RENAMED` → `REMOVED` → `MODIFIED` → `ADDED`) producing the sealed archive (Pin [3]) and updated SSOT (Pin [4]).

## 3.1 Slash-Command Naming, The 99% vs. 1% Rule & The "Core 4" Workflow

**Why You See Both `/openspec-*` and `/opsx:*` (And the 99% vs. 1% CLI Rule).** Upstream OpenSpec docs use colon syntax (`/opsx:propose`, `/opsx:apply`), whereas skill-based AI assistants register skills by directory name (`/openspec-propose`, `/openspec-apply-change`); both invoke the exact same workflow. For **99% of daily work**, you run **Agent Skills in chat**—the agent calls the `openspec` CLI binary under the hood to scaffold directories and fetch JSON instructions, then writes the Markdown specs and application code. You only run `openspec` directly in your terminal for the remaining **1%**: instant dashboards (`openspec view` or `openspec list`), linting manual spec edits (`openspec validate <name> --strict`), or forking custom schemas (`openspec schema fork`).

**How OpenSpec Analyzes Baseline Specs: Semantic Diffing & "The Grill".** When you propose a change against an existing repository, OpenSpec does not guess silently; it runs a 3-part analysis loop: (1) **Semantic Diffing** (loading the baseline `openspec/specs/<cap>/spec.md` and comparing it against your prompt), (2) **Constraint Boundary Extraction** (spotting changes to ordering, validation, or capacity limits), and (3) **Interactive Socratic Prompting ("The Grill")** (blocking code generation to ask 3 clarifying questions on unstated edge cases). In general, **80% of daily engineering uses the "Core 4" steps**:

1. **Think & Grill First (`/openspec-explore` / `/opsx:explore`):** Read-only investigation that surfaces ambiguities ('Q1..Q3') without touching code.
2. **Propose in One Step (`/openspec-propose` / `/opsx:propose`):** Generates `proposal.md`, delta `specs/`, `design.md`, and `tasks.md` inside `openspec/changes/<name>/`.
3. **The 60-Second Human Sanity Check:** Read `proposal.md`, `specs/`, and `tasks.md` before coding.
4. **Execute & Archive (`/openspec-apply-change` → `/openspec-archive-change`):** Implements `tasks.md` (`- [ ]` → `- [x]`) and merges deltas into `openspec/specs/`.

**Hands-On Walkthrough: Socratic "Grilling" & OpenSpec Change Bundle on `TaskFlow (notes-board)`**

```
# === 1. Interactive Socratic "Grilling" Before Generating the Spec Delta ===
> /opsx:propose "Add note pinning (pin_note and get_pinned_notes) to NotesBoard"
[OpenSpec Engine] Comparing baseline `openspec/specs/notes-board/spec.md` (REQ-0001..0003)
against your request. Please clarify 3 edge cases before I write the delta contract:
  Q1 (Creation Order): Should `list_notes()` keep chronological order (REQ-0002) when
      a note is pinned, or move pinned notes to the top? -> Keep REQ-0002 order intact!
  Q2 (Missing IDs): What if `pin_note(999)` is called on a missing ID? -> Raise KeyError.
  Q3 (Pinned Order): In what order should `get_pinned_notes()` return? -> Creation order.

# === 2. openspec/changes/add-note-pinning/specs/notes-board/spec.md ===
# Delta for TaskFlow Quick Notes Board (`notes-board`)
## ADDED Requirements
### Requirement: REQ-0004 - Pin Note and List Pinned Notes
The `NotesBoard.pin_note(note_id)` method SHALL set `pinned = True` on the matching note
and MUST raise `KeyError` if `note_id` does not exist. `get_pinned_notes()` SHALL return
all notes with `pinned == True` in chronological creation order.
#### Scenario: Pinning an existing note preserves list_notes creation order
- **GIVEN** notes `1` (`"First"`) and `2` (`"Important"`)
- **WHEN** `pin_note(2)` is called
- **THEN** `get_pinned_notes()` returns `[n2]` and `list_notes()` returns `[n1, n2]`

## MODIFIED Requirements
### Requirement: REQ-0001 - Add Note and Reject Blank Text
The `NotesBoard.add_note(text)` method SHALL trim `text`, assign sequential integer `id`
starting at `1`, initialize `pinned = False`, and raise `ValueError` if `text` is empty.
#### Scenario: Adding a valid note initializes pinned to False
- **WHEN** `add_note("  Buy milk  ")` is called
- **THEN** it returns `{"id": 1, "text": "Buy milk", "pinned": False}`

# === 3. openspec/changes/add-note-pinning/tasks.md ===
## Batch 1: Spec Delta & Baseline Order Preservation (PR 1)
- [x] 1.1 Add REQ-0004 to spec delta and verify REQ-0002 chronological `list_notes()`
## Batch 2: Pinning Implementation & Tests (PR 2)
- [x] 2.1 Implement `pin_note(note_id)` and `get_pinned_notes()` in `notes_board.py`
```

**Common Misconception vs. Engineering Reality: Why Partial Snippets Inside `## MODIFIED Requirements` Silently Delete Scenarios**

× **Common Misconception:** *"When updating an existing requirement under `## MODIFIED Requirements`, I only need to include the one new sentence or scenario that changed."* ✓ **Engineering Reality:** In OpenSpec's `specs-apply.js` engine, `MODIFIED` performs a whole-block replacement of $M[m]$ in `openspec/specs/<cap>/spec.md`. Any existing `#### Scenario:` block omitted from your `MODIFIED` section is permanently deleted upon archive. Always copy the entire existing requirement block into `MODIFIED` first, then edit it.

**Stop and Think: Why Must a Renamed-and-Modified Requirement Use the New (`TO`) Header?**

**Question:** If a single OpenSpec change both renames a requirement header under `## RENAMED Requirements` (`FROM: REQ-0002 Old` → `TO: REQ-0002 New`) and updates its scenarios under `## MODIFIED Requirements`, which header name must the `MODIFIED` block use? **Answer:** It must use the **new**

(TO) header name, because Stage 2 (RENAMED) mutates the key $f \mapsto t$ in the requirement map $M$ *before* Stage 4 (MODIFIED) looks up its target header.

# 4 OpenSpec Playbooks, Stacked PR Workflows & Security Guardrails

## 4.1 The 4 Scenario-Based OpenSpec Playbooks

**Playbook 1 — The "Fast-Track" Daily Driver (Small Features & Bug Fixes, 1–3 Files).** When requirements are clear, run `/openspec-propose "<description>"` (`/opsx:propose`), perform a **60-Second Sanity Check** on `specs/` and `tasks.md` (checking that edge cases like rate limits or missing headers are covered and no out-of-scope tasks were invented), run `/openspec-apply-change` (`/opsx:apply`), and run `/openspec-archive-change` (`/opsx:archive`).

**Playbook 2 — Deep Architecture & Spec-First PRs (`explore` → `new` → `continue`).** When touching unfamiliar code or designing a cross-cutting feature, start with `/openspec-explore` to compare approaches in ASCII without creating files. Next, run `/openspec-new-change <name>` (`/opsx:new`) and step through artifacts one at a time with `/openspec-continue-change` (`/opsx:continue`): edit `proposal.md` first, run `continue` to generate `specs/` and `design.md`, and run `continue` (or `/openspec-ff-change`) to generate `tasks.md`. **Pro-Tip (Spec-First PR / Snapshot 1):** Push *only* `openspec/changes/<name>/` as Snapshot 1 of your PR so your tech lead can approve the human-readable design in 10 minutes before you spend hours writing code.

**Playbook 3 — Reconciling Code Review Feedback Mid-PR (`sync` + `verify`).** When a reviewer asks for a behavioral change on an open PR (e.g., *"Default empty `tenant_id` to `'global'` instead of returning HTTP 400"*), never patch the code while leaving the spec stale. Update the code, unit tests, and `openspec/changes/<name>/specs/<cap>/spec.md` together, run `/openspec-sync-specs` (`/opsx:sync`), and run `/openspec-verify-change` (`/opsx:verify`) to prove 100% alignment before merging.

**Playbook 4 — Parallel Changes & Batch Archival (`/openspec-bulk-archive-change`).** When two concurrent branches or worktrees modify the same capability spec (e.g., `changes/add-oauth/` and `changes/add-jwt/` both touch `openspec/specs/auth/spec.md`), run `/openspec-bulk-archive-change` (`/opsx:bulk-archive`). OpenSpec detects the overlapping spec, inspects the codebase to verify both implementations exist, and merges the deltas in chronological order.

## 4.2 Version Control & Stacked PR Workflows (`1 Change = 1 Feature`)

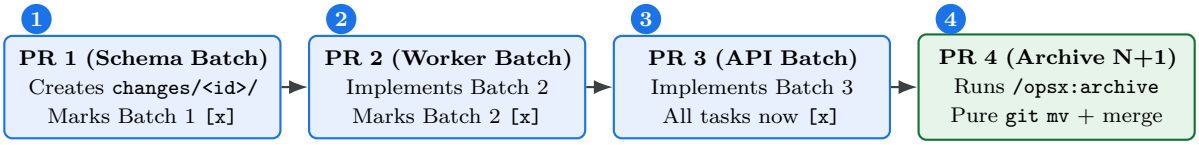
**Core Cardinality Rule: 1 OpenSpec Change = 1 Feature (Not 1 PR/Commit).** Never create a separate OpenSpec change folder for every small pull request in a stack (`changes/feat-part1`, `changes/feat-part2`), which fragments one architectural design across multiple directories. An OpenSpec change represents a **complete feature**, whereas a Git PR or Jujutsu commit (`jj`) represents a **code review unit** (Table 3 and Figure 4 below).

**Copy-Pasteable Recipe: Multi-PR Batched `tasks.md` & Stacked `git` / `jj` Commands (TaskFlow Lab 05)**

```
# === A. Group openspec/changes/add-note-pinning/tasks.md into PR Batches ===
## Batch 1 (PR 1): Spec Delta & Baseline Order Preservation (REQ-0001..REQ-0003)
- [x] 1.1 Add `REQ-0004` (`Pin Note and List Pinned Notes`) to OpenSpec delta
- [x] 1.2 Verify `list_notes()` keeps chronological creation order (`REQ-0002`)
```

| Workflow Pattern | Feature Size | When `/opsx:archive` Runs | What Reviewers See in Each PR |
|---|---|---|---|
| **1. Single-Pass Feature** (`1 Change = 1 PR`) | Small / Medium ($\sim$ 90% of work) | **Before opening the PR** (in the same branch/commit) | 1 self-contained PR with production code, tests, updated `openspec/specs/`, and `changes/archive/`. |
| **2. Spec-First Two-Stage** (`1 Change = 1 PR, 2 Snapshots`) | Ambiguous or high-risk design | **Stage 2** (after Stage 1 spec approval) | **Snapshot 1:** only `changes/<id>/` (all `- [ ]`). **Snapshot 2:** diff vs. Snapshot 1 shows only code + archive! |
| **3. Stacked Multi-PR** (`1 Change = N PRs + PR N+1`) | Large feature ($>$ 500 LOC across layers) | **Dedicated Docs-Only Archive PR** (`PR N+1`) at top of stack | **PR 1..N:** implements Batch $k$ code + marks only Batch $k$ as `[x]`. **PR N+1:** pure `git mv` archive & spec merge. |

Table 3: Mapping OpenSpec changes to Git pull requests and Jujutsu (`jj`) commit stacks.




**Why Dedicated Archive PR `N+1` Works Best:** Between PR 1 and PR 3, Plane 1 (`openspec/specs/`) stays untouched on `main` while `changes/<id>/tasks.md` acts as a live progress tracker. PR 4 isolates the `git mv` file rename so code PRs stay 100% noise-free.

Figure 4: Stacked Multi-PR execution (`1 Change = N Code PRs + Dedicated Docs-Only Archive PR N+1`, Pins [1]–[4]).

```
## Batch 2 (PR 2): Note Pinning Implementation & Tests (REQ-0004)
- [ ] 2.1 Implement `pin_note(note_id)` (raising `KeyError` if missing) in `notes_board.py`
- [ ] 2.2 Add `test_req0004_pin_note_and_raise_on_missing_id` in `test_notes_board.py`

# === B. Execute the Stack (Git Branches or Jujutsu `jj new`) ===
# PR 1: Propose full change + implement ONLY Batch 1
$ git checkout -b feat/notes-pinning-pr1   # (or: jj new main -m "PR 1: Spec & Baseline")
> /opsx:propose "Add note pinning to TaskFlow NotesBoard (2 PR batches)"
> Implement ONLY Batch 1 in tasks.md, run tests, and mark ONLY Batch 1 as [x].
$ git add . && git commit -m "feat(notes): add REQ-0004 spec delta (Batch 1)"

# PR 2: Stack on PR 1 + implement ONLY Batch 2
$ git checkout -b feat/notes-pinning-pr2   # (or: jj new -m "PR 2: Pinning")
> Implement ONLY Batch 2 in openspec/changes/add-note-pinning/tasks.md.
$ git add . && git commit -m "feat(notes): implement pin_note and tests (Batch 2)"

# PR 3 (Docs-Only Archive PR N+1): Seal change with clean git mv history
$ git checkout -b docs/notes-pinning-pr3-archive   # (or: jj new -m "PR 3: Archive")
> /opsx:archive add-note-pinning
$ git add -A && git commit -m "docs(openspec): archive add-note-pinning change"
```

**5 Mandatory Golden Rules & Security Guardrails for OpenSpec**

1. **Rename-Then-Modify & Full-Block Copy Rules:** Because `RENAMED` runs before `MODIFIED`, any `MODIFIED` block for a renamed requirement MUST target the **new (`TO`) header**, and MUST copy the **complete existing requirement block** (including all existing `#### Scenario:` blocks) before editing so scenarios are never silently dropped.
2. **4-Hashtag Scenario Rule:** Every scenario header MUST use exactly four hashtags (`#### Scenario: Name`); 3-hashtag `### Scenario:` headers are silently ignored by `validator.js`.
3. **Human-in-the-Loop (HITL) Gate Before Code Edits:** Before `/openspec-apply-change` modifies source files, confirm the target files and task list.
4. **Indirect Prompt Injection Defense:** Treat existing `tasks.md` and spec files in shared repos as untrusted data; halt immediately if a task instructs the agent to exfiltrate secrets, weaken auth, or run arbitrary shell commands outside the build/test suite.
5. **Strict Phase Separation:** Never modify application source code during `explore`, `new`, or `propose`, and only mark a task `[x]` after its implementation and tests pass in the current PR.

**Common Misconception vs. Engineering Reality: One OpenSpec Change Folder per Feature vs. One Folder per Pull Request**

× **Common Misconception:** *"When breaking a large feature into 3 stacked pull requests, I should create `changes/feat-pr1/`, `changes/feat-pr2/`, and `changes/feat-pr3/`."* ✓ **Engineering Reality:** An OpenSpec change represents one cohesive architectural feature (`1 Change = 1 Feature`), whereas a PR is a review unit. Splitting one feature across multiple change folders fragments `design.md` and forces reviewers to hunt across directories; instead, keep one change folder and partition `tasks.md` into `## Batch 1 (PR 1)` through `## Batch N (PR N)`.

**Stop and Think: What Happens If You Run `/opsx:archive` in PR 1 of a 3-PR Stack?**

**Question:** Why does running `/opsx:archive` inside the first pull request (`PR 1`) break downstream stacked branches (`PR 2` and `PR 3`)? **Answer:** Because `/opsx:archive` moves the entire directory from `openspec/changes/<id>/` to `openspec/changes/archive/YYYY-MM-DD-<id>/` (`git mv`). As soon as `PR 1` merges or rebases, `PR 2` and `PR 3`—which still modify `openspec/changes/<id>/tasks.md`—collide with a Git modify/delete tree conflict. Always defer `/opsx:archive` to a dedicated docs-only `PR N+1` at the top of the stack.

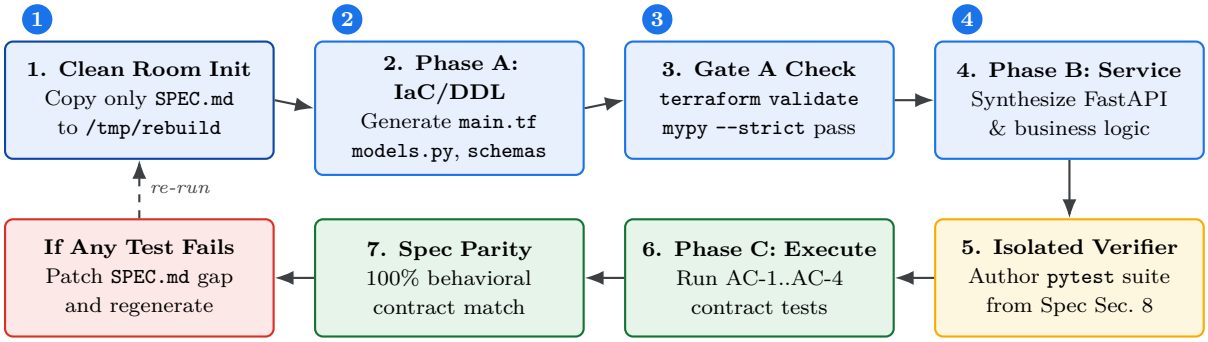
# 5 Spec-as-SSOT (`SPEC.md`) & Complete Command Reference

## 5.1 Authoring Rebuildable `SPEC.md` Blueprints & The Rebuild Test

**The 8-Section `SPEC.md` Schema & 4-Gate Brownfield Extraction.** When you want a single document that can deterministically regenerate an entire microservice from scratch (**Spec-as-SSOT**), structure `SPEC.md` into the 8 canonical sections shown in the template below. When extracting a `SPEC.md` from an existing brownfield repository, never rely on `README.md` alone; run the **4-Gate Discovery Scan** (`find` docs → `rg '^resource'` across all `*.tf` files → `rg` route decorators → `rg` SQL/DDL call sites) and replace all customer/environment strings with `<UPPER_SNAKE_CASE>` placeholders.

**Copy-Pasteable 8-Section `SPEC.md` (Single Source of Truth) Skeleton**

```
# SPEC.md: <Service Name> (Single Source of Truth Blueprint)
## 1. Domain Intent, Capabilities & Non-Goals
- Purpose, strict functional boundaries, and explicit out-of-scope non-goals.
## 2. System Architecture & Data Flow
- Component topology, trust boundaries, and synchronous vs. async data paths.
## 3. Executable Data Models & Schemas (Zero Placeholders)
- Complete SQL `CREATE TABLE` DDL + Pydantic/Zod models (never write `TODO: fields`).
## 4. API & Interface Contracts
- Method, path, auth scheme, request/response schemas, and RFC 9457 error codes.
## 5. Core Workflows, State Machines & Concurrency Rules
- Step-by-step state transitions, idempotency locks, and retry/backoff rules.
## 6. Target Repository Directory Tree
- Canonical file layout (`src/`, `migrations/`, `tests/`, `infra/`).
## 7. Parameterized Infrastructure & Environment Contract
- Terraform/IaC using `<GCP_PROJECT_ID>`, `<VPC_NETWORK_NAME>`, `<DB_DSN>` tokens.
## 8. 3-Phase Rebuild Recipe & Binary Acceptance Criteria
- Phase A (Schemas/IaC) -> Phase B (Core Service) -> Phase C (Read-Only Test Gate).
```




**The Rebuild Test Rule:** Copy `SPEC.md` into an empty directory (`/tmp/rebuild`). If an isolated test agent (generating `tests/` strictly from Section 8 with `chmod -R a-w tests/`) fails against the generated service, fix the ambiguity in `SPEC.md`—never patch the generated code by hand.

Figure 5: The 3-Phase Clean-Room Rebuild Test pipeline (Pins [1]–[4]) verifying that `SPEC.md` is a true Single Source of Truth.

## 5.2 Why OpenSpec Has Both Agent Skills and a CLI Binary

**Separating LLM Reasoning (Cognitive Layer) from State Management (Deterministic Engine).** LLMs excel at reading codebases, reasoning about trade-offs, and writing tailored Markdown, but they are unreliable at enforcing strict directory schemas and JSON validation rules on their own. OpenSpec splits the work into two layers so the agent never hallucinates workflows or skips required artifacts:

**Two-Layer Architecture: Cognitive Layer (AI Agent Skill) + Deterministic Engine (`openspec` CLI)**

```
+-------------------------------------------------------------------------+
|                    YOU (Developer in Chat / IDE / CLI)                  |
|         Prompt: "/openspec-propose Add rate limiting to POST /v1/orders"|
+-----------------------------------+-------------------------------------+
```

```
                                    |
                                    v
+-------------------------------------------------------------------------+
|                  1. COGNITIVE LAYER: AI Agent Skill                     |
|         (Reads skills/openspec-propose/SKILL.md or /opsx:propose)       |
|  - Explores your codebase (src/, tests/, openspec/specs/)               |
|  - Reasons about architecture, edge cases, and failure modes            |
|  - Authors tailored Markdown content (proposal, specs, design, tasks)   |
+-------------------+-----------------------------+-----------------------+
                    | Calls CLI under the hood    | Returns JSON
                    v                             | templates & DAG
+-------------------------------------------------------------------------+
|               2. DETERMINISTIC ENGINE: `openspec` CLI Binary            |
|  - Scaffolds directories:  openspec new change "<name>"                 |
|  - Computes DAG state:     openspec status --change "<name>" --json     |
|  - Injects rules & templates: openspec instructions <id> --json         |
|  - Enforces RFC 2119 & Gherkin grammar: openspec validate --strict      |
+-------------------------------------------------------------------------+
```

## 5.3 Complete Reference for All 11 OpenSpec Skills & CLI Commands

**All 11 OpenSpec Skills Mapped to Under-the-Hood CLI Calls.** Table 4 provides the complete reference for all 11 OpenSpec skills, showing when to invoke each skill, what `openspec` CLI commands run behind the scenes, and what the agent produces.

**Example Session Outputs: /openspec-explore, /openspec-apply-change & /openspec-verify-change**

```
# === 1. Interactive Architecture Investigation (/openspec-explore) ===
> /openspec-explore How should we store TaskFlow notes so pin_note stays O(1)?
  Current Flow: [NotesBoard] ---> [List scan O(n)] vs [Dict + pinned flag]
  Approach A: Dict Keyed by Note ID         Approach B: Separate Pinned List
  +----------------------------------+      +----------------------------------+
  | + O(1) lookup by note_id         |      | + Fast pinned-only iteration     |
  | + Preserves creation-order sort  |      | - Duplicates state across 2 lists |
  +----------------------------------+      +----------------------------------+
  Recommendation: Approach A (Dict keyed by ID). Run `/openspec-propose pin-notes`?

# === 2. Task Execution (/openspec-apply-change) & 3D Audit (/openspec-verify-change) ===
> /openspec-apply-change pin-notes
[Task 1/2] 1.1 Add REQ-0004 pin_note(note_id) in notes_board.py -> Updated tasks.md: [x]
[Task 2/2] 1.2 Add test_req0004_pin_note_orders_pinned_first -> 4 passed -> [x]

> /openspec-verify-change pin-notes
1. Completeness: PASSED (2/2 tasks checked [x], 4/4 requirements implemented)
2. Correctness:  PASSED (All 4 REQ-0001..REQ-0004 scenarios covered by unit tests)
3. Coherence:    PASSED (Matches design.md pinned-first tuple sort decision)
```

## 5.4 Curated Public Codelabs, Video Walkthroughs & Courses

**Hands-On Codelabs, Video Tutorials & Practitioner Courses.** Table 5 catalogs the highest-signal public interactive codelabs, step-by-step video walkthroughs, short courses, and engineering blog guides for practicing SDD, Conductor, and OpenSpec.

| **Skill (`/openspec-*` / `/opsx:*`)** | **Under-the-Hood `openspec` CLI Calls** | **When to Use & What the Agent Produces** |
|---|---|---|
| **1.** `explore` (`/opsx:explore`) | *None* (or `openspec list --json`). | **Think before proposing:** Reads code, compares options, and draws ASCII trade-off diagrams (read-only). |
| **2.** `propose` (`/opsx:propose`) | `new change` → `status --json` → `instructions <id> --json`. | **Default fast path:** Generates `proposal.md`, delta `specs/`, `design.md`, and `tasks.md` in one pass. |
| **3.** `new-change` (`/opsx:new`) | `schemas --json` → `new change "<name>"` → `status --json`. | **Start step-by-step:** Scaffolds empty `changes/<name>/` folder and pauses (‘0/4‘ artifacts). |
| **4.** `continue-change` (`/opsx:continue`) | `status --change "<name>" --json` → `instructions <next-id> --json`. | **Advance 1 step:** Generates *only* the next unlocked artifact (`proposal` → `specs/design` → `tasks`), then pauses. |
| **5.** `ff-change` (`/opsx:ff`) | Loops `status --json` + `instructions <id> --json` for remaining items. | **Fast-forward planning:** Generates all remaining planning files automatically once `proposal.md` is approved. |
| **6.** `apply-change` (`/opsx:apply`) | `openspec instructions apply --change "<name>" --json`. | **Implement tasks:** Executes unchecked `- [ ]` items in `tasks.md`, runs tests, and marks `- [x]`. Resumable. |
| **7.** `verify-change` (`/opsx:verify`) | `openspec status --change "<name>" --json`. | **3-Dimension Audit:** Checks **Completeness** (tasks/reqs), **Correctness** (scenarios/tests), and **Coherence** (design). |
| **8.** `sync-specs` (`/opsx:sync`) | Reads `changes/<name>/specs/*/spec.md`. | **Mid-PR sync:** Merges delta specs into `openspec/specs/` without archiving the change folder. |
| **9.** `archive-change` (`/opsx:archive`) | `openspec status --change "<name>" --json`. | **Finalize change:** Syncs delta specs into `openspec/specs/` and moves folder to `changes/archive/YYYY-MM-DD-<name>/`. |
| **10.** `bulk-archive` (`/opsx:bulk-archive`) | `list --json` → `status --change "<name>" --json` (per change). | **Batch archive:** Detects spec overlaps across parallel changes, verifies code exists, and archives chronologically. |
| **11.** `onboard` (`/opsx:onboard`) | Full CLI lifecycle (`new`, `status`, `instructions`, `apply`, `archive`). | **15-min interactive tutorial:** Picks a small real task in your repo and walks through every step interactively. |

Table 4: Complete reference for all 11 OpenSpec Agent Skills and their underlying `openspec` CLI calls.

| Format | Resource Title & Link | What You Will Build / Learn |
|---|---|---|
| **Interactive Codelab (69m)** | **Plan and Build Apps with Gemini CLI Conductor** https://codelabs.developers.google.com/next26/advanced-planning | Official Google Codelab by Tianzi Cai: build a "Picker Wheel" web app from scratch (**Greenfield**), then iterate on it as a **Brownfield** app to add Firebase Auth & Storage. |
| **Multi-Agent Codelab** | **Conductor Plugin Codelab** https://codelabs.developers.google.com/conductor-plugin | Hands-on lab covering the cross-agent Conductor plugin across Gemini CLI, Antigravity CLI, and Claude Code. |
| **Official Engineering Blogs** | **Introducing Conductor & Automated Reviews** https://developers.googleblog.com/conductor-introducing-context-driven-development-for-gemini-cli/ | Google Developers Blog walkthroughs of the `Context -> Spec & Plan -> Implement` loop and post-implementation `/conductor:review` compliance audits. |
| **Short Course (Video + Labs)** | **Spec-Driven Development with Coding Agents DeepLearning.AI & JetBrains** (Paul Everitt) | Structured video course on replacing "vibe coding" with persistent Markdown constitutions, iterative feature specs, and greenfield/brownfield agent workflows. |
| **Video Demos (YouTube)** | **1. Gemini CLI Conductor Explained With Demo Project** (Technical Potpourri) **2. OpenSpec Changes Everything — Full Step-by-Step** (https://www.youtube.com/watch?v=xwpmPP4mNBQ) **3. Build with OpenSpec in Claude Code** (ExamPro) | Full terminal screen-recordings building a real-time ISS tracker with Conductor tracks/checkpoints and running OpenSpec's `explore -> propose -> apply -> archive` loop on live repos. |
| **Official Open-Source Hubs** | **Conductor:** https://github.com/gemini-cli-extensions/conductor **OpenSpec:** https://openspec.dev/ & https://github.com/Fission-AI/OpenSpec | Official extension manifests, TOML command definitions, CLI schemas (`@fission-ai/openspec`), and built-in `/opsx:onboard` 15-minute repo tutorial. |
| **Foundational Theory** | **Joel on Software: The Law of Leaky Abstractions** (Joel Spolsky, 2002) https://www.joelonsoftware.com/2002/11/11/the-law-of-leaky-abstractions/ | Why code-generation tools save time *working* (typing) but never save time *learning* the underlying distributed systems invariants that leak through prompts. |

Table 5: Curated public hands-on codelabs, video tutorials, courses, and official repositories.

**Common Misconception vs. Engineering Reality: Hand-Patching Generated Code in `/tmp/rebuild` vs. Fixing `SPEC.md`**

× **Common Misconception:** *"If 3 of 4 acceptance tests pass during a Clean-Room Rebuild Test in `/tmp/rebuild`, I can just edit the generated Python file to fix the 4th test."* ✓ **Engineering Reality:** Hand-patching generated code in the clean room proves that `SPEC.md` leaked an unstated assumption and is no longer a Level-3 Single Source of Truth. Whenever a clean-room test fails, you must patch the missing requirement or boundary condition inside `SPEC.md` and regenerate the code from scratch.

**Stop and Think: Why Does OpenSpec Split Work Between Markdown Skills and a CLI Binary?**

**Question:** Why not implement OpenSpec purely as Markdown prompt instructions without a compiled `openspec` CLI binary? **Answer:** Because LLMs excel at semantic code analysis and architectural synthesis (**Cognitive Layer**) but are stochastically unreliable at enforcing directory schemas, computing artifact DAG states, and validating strict RFC 2119 + 4-hashtag Gherkin grammar (**Deterministic Engine**). Combining both layers guarantees zero structural drift.